

# Bayesian zero-order TSK fuzzy system modeling

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## ABSTRACT

Different from the existing TSK fuzzy system modeling methods, a novel zero-order TSK fuzzy modeling method called Bayesian zero-order TSK fuzzy system (B-ZTSK-FS) is proposed from the perspective of Bayesian inference in this paper. The proposed method B-ZTSK-FS constructs zero-order TSK fuzzy system by using the maximum a posteriori (MAP) framework to maximize the corresponding posteriori probability. First, a joint likelihood model about zero-order TSK fuzzy system is defined to derive a new objective function which can assure that both antecedents and consequents of fuzzy rules rather than only their antecedents of the most existing TSK fuzzy systems become interpretable. The defined likelihood model is composed of three aspects: clustering on the training set for antecedents of fuzzy rules, the least squares (LS) error for consequent parameters of fuzzy rules, and a Dirichlet prior distribution for fuzzy cluster memberships which is considered to not only automatically match the “sum-to-one” constraints on fuzzy cluster memberships, but also make the proposed method B-ZTSK-FS scalable for large-scale datasets by appropriately setting the Dirichlet index. This likelihood model indeed indicates that antecedent and consequent parameters of fuzzy rules can be linguistically interpreted and simultaneously optimized by the proposed method B-ZTSK-FS which is based on the MAP framework with the iterative sampling algorithm, which in fact implies that fuzziness and probability can co-jointly work for TSK fuzzy system modeling in a collaborative rather than repulsive way. Finally, experimental results on 28 synthetic and real-world datasets are reported to demonstrate the effectiveness of the proposed method B-ZTSK-FS in the sense of approximation accuracy, interpretability and scalability.

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## 1. Introduction

To solve the regression estimation problems effectively using input–output data collected from a real process, all kinds of data-driven models have been proposed. Well-known regression models include fuzzy systems (FSs) [1,2], support vector regression (SVRs) [3,4], neural networks (NNs) [5,6], and regression trees [7,8], multivariate adaptive regression splines (MARS) [9,10]. For each model, there are situations for which it is particularly well suited. For example, the regression trees and MARS are capable of processing irrelevant trees and handling missing data, respectively. However, their predictive power is usually not as good as that of SVRs and FSs [11,12]. NNs and SVRs are based on minimizing the empirical risk and the structural risk, respectively. Both models operate as a black box. Compared with NNs and SVRs, one distinctive advantage

of FSs is transparency, interpretability in how they comprise IF–THEN fuzzy rules. Besides, various FSs have been proved to be universal approximators in [13–18]. More detailedly, the universal approximation theorem holds true for the static fuzzy models with trapezoid-shaped membership functions in [13,14]. This theorem is extended to be true for both the dynamic fuzzy models and the static fuzzy models in [15,16]. An approach to robust controller design for general nonlinear systems based on FSs as universal approximators is developed in [17], and a class of generalized T-S FSs with control variables included in the premise vector is also proved to be universal approximators in [18]. Thus, this paper mainly focuses on the proposal of a new zero-order fuzzy system modeling method for regression estimation to improve the performance of fuzzy system in the sense of approximation accuracy, interpretability and scalability. As is known to us, classical fuzzy system models include the Mamdani–Larsen models [19], Takagi–Sugeno–Kang (TSK) models [20], and generalized fuzzy models [21]. At present, various TSK models can be generally categorized into three types. The first one is called zero-order TSK fuzzy system [20] in which a constant consequent part of each fuzzy

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rule is used. The second one deals with a linear model [20] used in each consequent part of each fuzzy rule, and the third one is often called higher order TSK model in which a higher order model [20] is adopted in each consequent part of each fuzzy rule. Due to its simplicity and universal approximation capability [1,17,18], here we prefer zero-order TSK fuzzy system and consider how to develop its learning algorithm from a new perspective.

At present, a lot of fuzzy system modeling strategies have been proposed, which can be roughly categorized into two categories.

- 1) The first category only tunes the consequent parameters of fuzzy rules. In this case, the rule base is fixed and the membership functions of the antecedents of fuzzy rules have been determined in advance through hard or fuzzy clustering methods to assure the interpretability of the premises of fuzzy rules. The least squares (LS) algorithm [22] and backpropagation (BP) algorithm [23] are two most typical examples among these methods for tuning the consequent parameters.
- 2) The second category deals with simultaneous learning for antecedents and consequent parameters of fuzzy rules in fuzzy systems. In fact, it is also a hotspot of current researches on fuzzy system modeling [24]. Fuzzy neural networks (FNNs) or neural-network-based fuzzy systems (NNFSs) [12,25], and evolutionary fuzzy systems or Genetic Fuzzy Systems (GFSs) [26,27] are two most popular fuzzy system modeling methodologies for achieving simultaneous learning for antecedents and consequent parameters of fuzzy rules in fuzzy systems. The distinctive advantage of this type is its high learning accuracy with a small/medium-sized dataset. However, the learning algorithms of this category usually accompany with slow convergence, overfitting phenomenon and lack of the interpretability of fuzzy rules for datasets with a large number of attributes and/or instances.

Since the first category of fuzzy system modeling strategies limits the flexibility of both antecedent parameter learning and consequent parameter learning of fuzzy systems, we focus on the second category in this study. Except for neural or evolutionary learning methods, the methods that can simultaneously tune antecedent and consequent parameters of fuzzy rules remain scarce. On the other hand, due to the prevalence of both fuzzy and probabilistic thinking, a group of researchers have used them in some combinational ways. In particular, in the community of machine learning, Bayesian inference is particularly important in data analysis, which has been popular in undertaking parameters optimization and inference tasks, due to its statistical inference ability [28]. Representative methods among these combinations include, for example, fuzzy Bayesian inference (FBI) [29], proposed by Yang who introduced a fuzzy membership function for achieving Bayesian inference under conditions of uncertainty, and Bayesian fuzzy clustering (BFC) [30], proposed by Glenn who introduced Bayesian techniques for fuzzy clustering to estimate the number of clusters by means of a prior approach. Other works have also combined fuzzy concepts with probabilistic techniques [31,32], or used fuzzy and probabilistic methods in the same system [33,34]. However, up to date, to the best of our knowledge, the study that directly deals with fuzzy system modeling using Bayesian inference framework for simultaneous antecedent and consequent parameters learning of fuzzy rules has not been reported.

Inspired by simultaneous learning for both antecedent and consequent parameters of fuzzy rules and leveraging the outstanding advantages of Bayesian inference framework, the objective of this paper is to design a new zero-order TSK fuzzy system modeling method based on Bayesian inference framework such that it can assure both antecedent parameters and consequent parameters of fuzzy rules become interpretable and simultaneously tunable

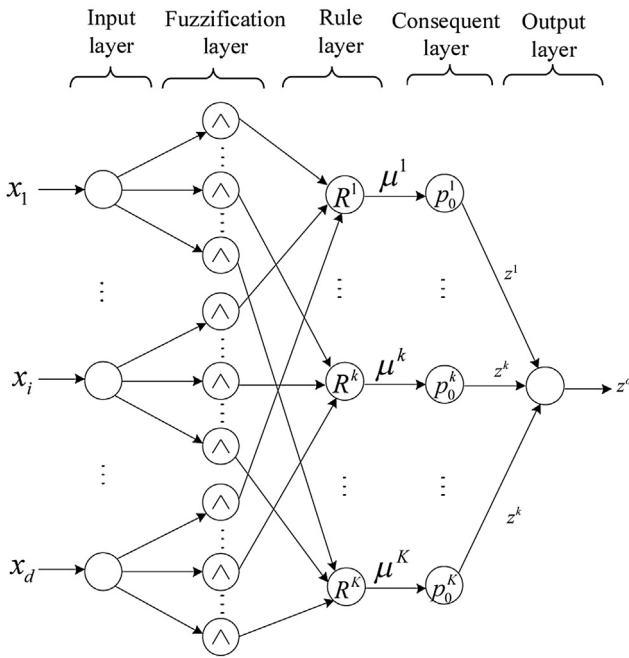
through the proposed iterative sampling algorithm. In view of its accuracy and interpretability, TSK fuzzy system is chosen to be incorporated with Bayesian inference framework and accordingly the Bayesian zero-order TSK fuzzy system (B-ZTSK-FS) modeling method will be proposed in this study. In the proposed method B-ZTSK-FS, by defining a data likelihood distribution about zero-order TSK fuzzy system, a Dirichlet prior distribution about the fuzzy cluster memberships used in the fuzzy rules and a special Gaussian distribution about cluster centers for antecedent and consequent parts of the fuzzy rules, a joint likelihood model about zero-order TSK fuzzy system is accordingly built to derive a new objective function that integrates the objective function of the fuzzy c-means (FCM) [35] with the least square (LS) error [22], and then the induced fuzzy rules of zero-order TSK fuzzy system are learnt through the proposed iterative sampling algorithm. The experimental results on 28 synthetic and real-world datasets show that the proposed modeling method B-ZTSK-FS for zero-order TSK fuzzy system provides the improved performance over the comparison methods. The major contributions of this study are as follows.

- 1) A joint likelihood model is defined for zero-order TSK fuzzy system modeling. Accordingly, a new fuzzy system modeling method B-ZTSK-FS is proposed, which can make both antecedent parameters and consequent parameters of fuzzy rules become interpretable and simultaneously tunable, which indeed demonstrates the distinctive advantage of the proposed modeling method B-ZTSK-FS over the existing TSK fuzzy system modeling methods in which only antecedent parameters of fuzzy rules are assured to be interpretable through using hard or fuzzy clustering methods.
- 2) Based on Bayesian inference framework with a Dirichlet prior distribution assumption about fuzzy cluster memberships, an iterative sampling algorithm is developed to find the maximum a posteriori (MAP) value of parameters involved. What is more, this sampling algorithm together with an appropriately-set Dirichlet index can realize a fast construction of zero-order TSK fuzzy system for a large-scale dataset. To our knowledge, it is the first time that we pull together Bayesian inference and TSK fuzzy system modeling in such a way. And the work indeed provides another encouraging fuzzy system modeling method in which fuzziness and probability can co-jointly work as collaborators rather than rivals.
- 3) The proposed modeling method B-ZTSK-FS for zero-order TSK fuzzy system is experimentally demonstrated that it can provide improved performance over the comparison methods in the sense of the approximation accuracy, interpretability and scalability. Theoretically, in terms of the theoretical claims about the MCMC methods in [36–38], as long as the number of iterations is enough, the global optima solution (as opposed to local optima solution found with the standard FCM and k-means algorithms) can be estimated in B-ZTSK-FS, because the proposed modeling method makes use of MCMC sampling techniques.

The content of this paper is organized as follows. In Section 2, the classical zero-order TSK fuzzy system and its neuro-fuzzy learning are introduced in brevity. In Section 3, a new modeling method B-ZTSK-FS for zero-order TSK fuzzy system is proposed from the perspective of Bayesian inference framework with a Dirichlet prior distribution assumption about fuzzy cluster memberships. In terms of the proposed method, a joint likelihood model and its learning based on iterative sampling are also discussed in this section. Section 4 illustrates the power of the proposed modeling method through reporting the experimental results of both the proposed modeling method and the comparison methods on 28 synthetic and real-world datasets. Finally, Section 5 draws the conclusions and directs the future work.

**Table 1**  
Mathematical notations used in the paper.

| Notations           | Explanations                                                                      |
|---------------------|-----------------------------------------------------------------------------------|
| $d$                 | The dimensionality of data objects, or the number of the input attributes         |
| $K$                 | The number of fuzzy rules, or the number of clusters                              |
| $N$                 | The number of all the data objects                                                |
| $\mathbf{x}_j$      | The $j$ th input data object, $j = 1, 2, \dots, N$                                |
| $x_i$               | The $i$ th input attribute of a data object, $i = 1, 2, \dots, d$                 |
| $A_i^k$             | The $i$ th fuzzy set in the $k$ th fuzzy rule, $k = 1, 2, \dots, K$               |
| $f^k(\mathbf{x})$   | The output of the $k$ th fuzzy rule                                               |
| $y_i^k, \delta_i^k$ | The mean and standard deviation of Gaussian fuzzy membership function             |
| $\mu^k(\mathbf{x})$ | the firing strength of the $k$ th fuzzy rule                                      |
| $R^k$               | the $k$ th fuzzy rule                                                             |
| $z$                 | The desired output of the input data object                                       |
| $z^o$               | The TSK fuzzy system output of the input data object                              |
| $\mathbf{x}_g$      | The global vector constructed from all $\mu^k(\mathbf{x})$ , $k = 1, 2, \dots, K$ |
| $p_0^k$             | The constant consequent parameter of the $k$ th fuzzy rule                        |
| $\mathbf{p}_0$      | The global vector constructed from all $p_0^k$ , $k = 1, 2, \dots, K$             |



**Fig. 1.** Neural structure of zero-order TSK fuzzy system.

Main mathematical notations used in this paper are listed in Table 1 to enhance readability.

## 2. Zero-order TSK fuzzy system and its neural learning

In this section, we will briefly introduce zero-order TSK fuzzy system and its neural structure. The knowledge base of zero-order TSK fuzzy system consists of  $K$  fuzzy IF–THEN rules. That is to say, the  $k$ th fuzzy rule, denoted as  $R^k$ , is expressed in the following form,

$$R^k: \text{IF } x_1 \text{ is } A_1^k \wedge x_2 \text{ is } A_2^k \wedge \dots \wedge x_d \text{ is } A_d^k \\ \text{THEN } f^k(\mathbf{x}) = p_0^k, k = 1, 2, \dots, K. \quad (1)$$

where  $A_i^k$  is a fuzzy set subscribed by the  $i$ th input attribute  $x_i$  for the  $k$ th fuzzy rule, and  $\wedge$  is a fuzzy conjunction operator. Each rule is premised on the input data object  $\mathbf{x} = (x_1, x_2, \dots, x_d)^T$ , and maps the fuzzy sets in the input space  $A^k \subset R^d$  to a varying singleton denoted by  $f^k(\mathbf{x})$ , and  $p_0^k$  is the crisp consequent parameter for the  $k$ th fuzzy rule. Generally, zero-order TSK fuzzy system can be trained by using the following five-layer neural structure in Fig. 1.

A detailed description of the functionality of each layer is given as follows:

- 1) Input layer: No calculation is carried out in this layer. Each node in this layer corresponds to an input attribute and directly transmits the input data object  $\mathbf{x} = (x_1, x_2, \dots, x_d)^T$  to the next layer.
- 2) Fuzzification layer: Each node in this layer corresponds to a fuzzy membership function. For input attribute  $x_i$ , the following Gaussian membership function is adopted,

$$\mu_{A_i^k}(x_i) = \exp \left\{ \frac{-(x_i - y_i^k)^2}{2\delta_i^k} \right\}, i = 1, 2, \dots, d \text{ and } k = 1, 2, \dots, K. \quad (2)$$

where  $y_i^k$  and  $\delta_i^k$  denote the corresponding mean and deviation, respectively, of the Gaussian membership function. The number of nodes in this layer is equal to  $K \times d$ .

- 3) Rule layer: The number of nodes in this layer actually is equivalent to the number of fuzzy rules. Each node performs a T-norm operation on the outputs of fuzzification layer using an algebraic product operation to obtain a firing strength or degree of match of the input to the fuzzy sets,  $\mu^k(\mathbf{x})$ . Here, the firing strength  $\mu^k(\mathbf{x})$  is obtained by,

$$\mu^k(\mathbf{x}) = \prod_{i=1}^d \mu_{A_i^k}(x_i) = \exp \left\{ -\sum_{i=1}^d \left( \frac{(x_i - y_i^k)^2}{2\delta_i^k} \right) \right\}, \quad (3)$$

where  $y_i^k$  and  $\delta_i^k$  will be explained later.

- 4) Consequent layer: In terms of the corresponding firing strength  $\mu^k(\mathbf{x})$  from the rule layer, the output function of node  $k$  can be formulated as,

$$z^k(\mathbf{x}) = \mu^k(\mathbf{x}) / \sum_{k'=1}^K \mu^{k'}(\mathbf{x}) \cdot f^k(\mathbf{x}) = \tilde{\mu}^k(\mathbf{x}) \cdot p_0^k \quad (4)$$

where  $\tilde{\mu}^k = \mu^k(\mathbf{x}) / \sum_{k'=1}^K \mu^{k'}(\mathbf{x})$ .

- 5) Output layer: This layer consists of a single output node which integrates the multiple output attributes from each node in consequent layer and calculates a simple weighted sum. The fuzzy system output can then be written as,

$$z^o = \sum_{k=1}^K z^k(\mathbf{x}) \\ = \sum_{k=1}^K \left( \tilde{\mu}^k(\mathbf{x}) \cdot p_0^k \right). \quad (5)$$

In order to find the parameters in the above neural structure of zero-order TSK fuzzy system, we may take one of the two learning methods below.

- 1) Learning antecedent parameters and consequent parameters separately. In this method, the parameters  $y_i^k$  and  $\delta_i^k$  in Eq. (3) can be estimated by clustering techniques or other partition methods [39]. For example, with the FCM algorithm [35],  $y_i^k$  and  $\delta_i^k$  can be estimated [40] as follows,

$$y_i^k = \frac{\sum_{j=1}^N u_{jk} x_{ji}}{\sum_{j=1}^N u_{jk}}, \quad (6)$$

$$\delta_i^k = \frac{h \sum_{j=1}^N u_{jk} (x_{ji} - y_i^k)^2}{\sum_{j=1}^N u_{jk}}, \quad (7)$$

where  $u_{jk}$  denotes the fuzzy cluster membership of the  $j$ th input data object  $\mathbf{x}_j = (x_{j1}, x_{j2}, \dots, x_{jd})^T$  belonging to the  $k$ th cluster obtained by the FCM algorithm [35], and  $h$  is a scale parameter and can be adjusted manually.

After the premises of fuzzy rules in zero-order TSK fuzzy system are determined in advance according to Eqs. (6) and (7), we may define the following equations,

$$\mathbf{x}_g = \left( \mu^1(\mathbf{x}), \mu^2(\mathbf{x}), \dots, \mu^K(\mathbf{x}) \right)^T, \quad (8a)$$

$$\mathbf{p}_0 = (p_0^1, p_0^2, \dots, p_0^K)^T, \quad (8b)$$

then Eq. (5) can be formulated as the following linear regression problem [41],

$$z^o = \mathbf{p}_0^T \mathbf{x}_g. \quad (9)$$

Thus, the problem of consequent parameter learning of zero-order TSK fuzzy system can be transformed into the learning of the parameters in the corresponding linear regression model [40,42].

2) *Learning antecedent parameters and consequent parameters simultaneously.* In this method, an error function is usually first defined as follows:

$$\begin{aligned} \min_{y_i^k, \delta_i^k, p_0^k} E &= \sum_{j=1}^N (z_j^o - z_j)^2 \\ &= \sum_{j=1}^N \left( \sum_{k=1}^K \left( \tilde{\mu}^k(\mathbf{x}) \cdot p_0^k \right) - z_j \right)^2. \end{aligned} \quad (10)$$

The antecedent and consequent parameters  $y_i^k, \delta_i^k$  and  $p_0^k$  can be determined by the commonly-used gradient descent strategy. That is to say, we can derive the following update equations about these parameters,

$$y_i^k(t+1) = y_i^k(t) - \eta \frac{\partial E}{\partial y_i^k}, \quad (11a)$$

$$\delta_i^k(t+1) = \delta_i^k(t) - \eta \frac{\partial E}{\partial \delta_i^k}, \quad (11b)$$

$$p_0^k(t+1) = p_0^k(t) - \eta \frac{\partial E}{\partial p_0^k}, \quad (11c)$$

where  $\eta$  is the learning rate,  $t$  is the time step. The entire description of the corresponding learning algorithm based on Eq. (11) can be found in [43].

Here let us give the remarks about these two learning methods. As for a training dataset, with the help of hard or fuzzy clustering, the first learning method can construct a zero-order TSK fuzzy system in which each antecedent part of each fuzzy rule has interpretability. However, since it does not consider the intrinsic link between the distributions of both input data and output data, the obtained interpretability only for the premise of each fuzzy rule may not be good enough to make the consequent parameters of each fuzzy rule interpretable. Another shortcoming of the first learning method is its non-scalability for a large-scale training dataset. The second learning method often has very slow convergence and hence becomes also infeasible for a large-scale dataset. What is more, due to simultaneous learning for antecedent and consequent parameters of fuzzy rules, the interpretability for antecedent parameters can not be generally expected in the second

learning method. Therefore, it is worthwhile for us to develop the improved learning method such that it is effective for zero-order TSK fuzzy system modeling in the sense of approximation accuracy, interpretability and scalability.

### 3. The proposed modeling method B-ZTSK-FS

In this section, we will state a novel zero-order TSK fuzzy system modeling method B-ZTSK-FS by maximizing the proposed joint likelihood of both the training dataset and the antecedent and consequent parameters of zero-order TSK fuzzy system based on Bayesian inference framework and the maximum a priori (MAP) principle. The joint-likelihood-based objective function, adopted in the proposed method B-ZTSK-FS, attempts to find the maximal posterior probability of the antecedent and consequent parameters of fuzzy rules. Due to the introduction of a Dirichlet prior distribution assumption about fuzzy cluster memberships in the joint-likelihood-based objective function, the proposed modeling method B-ZTSK-FS for zero-order TSK fuzzy system can be implemented effectively by using Metropolis-Hastings iterative sampling technique [22].

#### 3.1. The proposed objective function

Suppose there is an input-output training dataset,

$$D = \{(\mathbf{x}_1, z_1), (\mathbf{x}_2, z_2), \dots, (\mathbf{x}_N, z_N)\}, \quad (12)$$

where  $N$  is the number of the data objects,  $\mathbf{x}_j \in R^d$  is the  $j$ th input data object, and  $z_j \in R$  is the  $j$ th desired output,  $j = 1, 2, \dots, N$ .

In terms of Bayesian inference framework and zero-order TSK fuzzy system, we first define a data likelihood distribution as follows:

$$\begin{aligned} p(\mathbf{X}, \mathbf{Z} | \mathbf{U}, \mathbf{Y}, \mathbf{p}_0) &= \prod_{j=1}^N p(\mathbf{x}_j, z_j | \mathbf{u}_j, \mathbf{Y}, \mathbf{p}_0) \\ &\propto \prod_{j=1}^N \exp \left\{ -\frac{1}{2} \sum_{k=1}^K u_{jk}^m \| (\mathbf{x}_j^T, z_j) - (\mathbf{y}_k^T, p_0^k) \|^2 \right\} \\ &\propto \exp \left\{ -\frac{1}{2} \sum_{j=1}^N \sum_{k=1}^K u_{jk}^m \| (\mathbf{x}_j^T, z_j) - (\mathbf{y}_k^T, p_0^k) \|^2 \right\}, \end{aligned} \quad (13)$$

in which  $N$  is the number of data objects,  $K$  is the number of clusters or fuzzy rules,  $u_{jk}$  is the fuzzy cluster membership of data object  $\mathbf{x}_j$  belonging to cluster  $k$ ,  $m$  is the fuzzy index. The input dataset  $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)^T$ , where  $\mathbf{x}_j = (x_{j1}, x_{j2}, \dots, x_{jd})^T$ , and the corresponding output vector  $\mathbf{Z} = (z_1, z_2, \dots, z_N)^T$ , the membership matrix  $\mathbf{U}$  with dimensions  $N \times K$  where the element at row  $j$ , column  $k$  is  $u_{jk}$ , and the  $K \times (d+1)$  cluster center matrix  $(\mathbf{Y}, \mathbf{p}_0) = ((\mathbf{y}_1^T, p_0^1)^T, (\mathbf{y}_2^T, p_0^2)^T, \dots, (\mathbf{y}_K^T, p_0^K)^T)^T$  of input-output data objects, where  $\mathbf{Y} = (\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_K)^T$ ,  $\mathbf{y}_k = (y_1^k, y_2^k, \dots, y_d^k)^T$  is a cluster center for antecedent part of the  $k$ th fuzzy rule,  $y_i^k, i = 1, 2, \dots, d$  is viewed as the mean of the Gaussian membership functions in Eq. (6) for zero-order TSK fuzzy system, and  $\mathbf{p}_0 = (p_0^1, p_0^2, \dots, p_0^K)^T$  in which  $p_0^k$  is the same as in the consequent part of the  $k$ th fuzzy rule (see Eq. (4)).

Next, after considering the “sum-to-one” constraints on the fuzzy cluster memberships, we take a Dirichlet prior distribution



for the fuzzy cluster memberships which can automatically match these “sum-to-one” constraints,

$$p(\mathbf{U}|\mathbf{Y}, \mathbf{p}_0) = \prod_{j=1}^N p(\mathbf{u}_j|\mathbf{Y}, \mathbf{p}_0) \propto \prod_{j=1}^N \text{Dirichlet}(\mathbf{u}_j|\boldsymbol{\alpha}), \quad (14)$$

in which  $\text{Dirichlet}(\mathbf{u}_j|\boldsymbol{\alpha})$  is a Dirichlet likelihood parameterized by the vector  $\boldsymbol{\alpha}$  defined as [30],

$$\text{Dirichlet}(\mathbf{u}_j|\boldsymbol{\alpha}) = \frac{\Gamma(\sum_{k=1}^K \alpha_k)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K u_{jk}^{\alpha_k-1}, \quad (15)$$

where  $u_{jk} \geq 0$  is the corresponding fuzzy cluster membership.  $\alpha_k > 0$  is the Dirichlet index.  $\sum_{k=1}^K u_{jk} = 1$  for all  $j = 1, 2, \dots, N$ .

Finally, since zero-order TSK fuzzy system can realize the regression between input data and output data, we may consider a special distribution on cluster centers for antecedent and consequent parts of the fuzzy rules as follows:

$$p(\mathbf{Y}, \mathbf{p}_0) \propto \prod_{k=1}^K \exp \left\{ -\frac{1}{2\sigma^2} \sum_{j=1}^N \|z_j^0 - z_j\|^2 \right\} \propto \exp \left\{ -\frac{1}{2\sigma^2} \sum_{k=1}^K \sum_{j=1}^N \left\| \sum_{k=1}^K (\tilde{\mu}^k(\mathbf{x}_j) \cdot p_0^k) - z_j \right\|^2 \right\} \quad (16)$$

in which  $\sigma$  is the standard deviation,  $z_j$  is mean, i.e. the desired output of the  $j$ th input data object, and  $z_j^0$  denotes the  $j$ th output of zero-order TSK fuzzy system for the  $j$ th input data object. Therefore, we have the joint likelihood of the data and parameters of zero-order TSK fuzzy system as follows:

$$p(\mathbf{X}, \mathbf{Z}, \mathbf{U}, \mathbf{Y}, \mathbf{p}_0) = p(\mathbf{X}, \mathbf{Z}|\mathbf{U}, \mathbf{Y}, \mathbf{p}_0)p(\mathbf{U}|\mathbf{Y}, \mathbf{p}_0)p(\mathbf{Y}, \mathbf{p}_0) \propto \exp \left\{ -\frac{1}{2} \sum_{j=1}^N \sum_{k=1}^K u_{jk}^m \left\| (\mathbf{x}_j^T, z_j) - (\mathbf{y}_k^T, p_0^k) \right\|^2 \right\} \times \left( \prod_{j=1}^N \prod_{k=1}^K u_{jk}^{\alpha_k-1} \right) \times \exp \left\{ -\frac{1}{2\sigma^2} \sum_{k=1}^K \sum_{j=1}^N \left\| \sum_{k=1}^K (\tilde{\mu}^k(\mathbf{x}_j) \cdot p_0^k) - z_j \right\|^2 \right\}, \quad (17)$$

which is proportional to the posterior probability distribution of the parameters,  $p(\mathbf{U}, \mathbf{Y}, \mathbf{p}_0|\mathbf{X}, \mathbf{Z}) \propto p(\mathbf{X}, \mathbf{Z}, \mathbf{U}, \mathbf{Y}, \mathbf{p}_0)$ . In terms of the MAP principle, by taking  $\sigma$  as 1 for simplicity, we can formulate the objective function of the joint likelihood model as the negative of its logarithm, i.e.,

$$J(\mathbf{X}, \mathbf{Z}, \mathbf{U}, \mathbf{Y}, \mathbf{p}_0) = \sum_{j=1}^N \sum_{k=1}^K u_{jk}^m \left\| (\mathbf{x}_j^T, z_j) - (\mathbf{y}_k^T, p_0^k) \right\|^2 - 2 \sum_{j=1}^N \sum_{k=1}^K (\alpha_k - 1) \log u_{jk} + \sum_{k=1}^K \sum_{j=1}^N \left\| \sum_{k=1}^K (\tilde{\mu}^k(\mathbf{x}_j) \cdot p_0^k) - z_j \right\|^2, \quad (18)$$

where a factor of 2 has been multiplied through to simplify.

**Remark 1.** We can draw the following conclusions 1), 2), and 3) from Eq. (18):

1) The first term in  $J(\mathbf{X}, \mathbf{Z}, \mathbf{U}, \mathbf{Y}, \mathbf{p}_0)$  is exactly the same as the objective function of FCM [35], which can guarantee to obtain the optimal cluster centers and fuzzy cluster memberships of each

input-output data objects. Due to the fact that the obtained cluster centers contain both antecedent and consequent parameters of fuzzy rules, we can explain one fuzzy rule from its overall perspective rather than only its antecedent part. Fuzzy partitions only for the input data can help us assign the whole antecedent part of each fuzzy rule with certain linguistic explanation, however, they can not give any hint for explaining consequent parameters in each fuzzy rule. In terms of the first term in  $J(\mathbf{X}, \mathbf{Z}, \mathbf{U}, \mathbf{Y}, \mathbf{p}_0)$ , the obtained cluster centers can be used to linguistically interpret both the whole antecedent part and the constant consequent part of each fuzzy rule, which is never done in the existing zero-order TSK fuzzy system modeling methods. In other words, the proposed objective function in fact induces more interpretability for the constructed zero-order TSK fuzzy system. In [44], two endeavors are contributed to make the consequent counterparts of fuzzy rules become interpretable, however, in contrast to fuzzy rules here, they use the so-called centralized fuzzy rules which are in a complicated form and hence may result in the performance degradation. Therefore  $J(\mathbf{X}, \mathbf{Z}, \mathbf{U}, \mathbf{Y}, \mathbf{p}_0)$  can help us give more interpretability for fuzzy rules.

2) The third term denotes the most popular east square (LS) criterion function in essence, i.e.,

$$\min_{\mathbf{p}_0} E = \sum_{j=1}^N (z_j^0 - z_j)^2. \quad (19)$$

The LS solution to the consequent parameters is generally through differentiation of the criterion function in Eq. (19) to yield the well-known pseudo-inverse solution, however, its computational complexity is expensive, which usually is  $O(N^3)$ . Instead, we figure out the LS solution by the MAP framework with the iterative sampling algorithm, in which the optimal consequent parameters can be guaranteed.

3) The second term in fact is derived by the Dirichlet likelihood defined in Eq. (15). The introduction of the Dirichlet prior distribution indeed produces two merits: 1) It can well express the positivity and “sum-to-one” constraints on the fuzzy cluster memberships. 2) Its sampling characteristic with different  $\boldsymbol{\alpha}$  values can make the proposed zero-order TSK fuzzy system modeling method B-ZTSK-FS become scalable for large-scale training datasets. In order to demonstrate the sampling characteristic of Dirichlet distribution, Table 2 gives three illustrative examples. We can readily observe that when  $\boldsymbol{\alpha}$  tends towards  $\mathbf{0}$  (e.g.,  $\boldsymbol{\alpha} = 0.001 \times \mathbf{1}_K$ , where  $\mathbf{1}_K$  is the  $K$  dimensional unit vector), fuzzy membership values in  $\mathbf{u}$  tend towards the extreme sampling (i.e., only one element being 1 and other elements being zero in  $\mathbf{u}$ ). When  $\boldsymbol{\alpha} = 0.01 \times \mathbf{1}_K$ , a little more non-zero fuzzy membership values in  $\mathbf{u}$  appear, which actually implies the sampling rate about non-zero fuzzy membership values becomes a little bigger. When  $\boldsymbol{\alpha} = 0.1 \times \mathbf{1}_K$ , much more non-zero fuzzy membership values in  $\mathbf{u}$  appear, which implies that much more computations will be involved in Eq. (17), due to the existence of much more non-zero fuzzy membership values in  $\mathbf{u}$ . Therefore, such a distinctive sampling characteristic of Dirichlet distribution with appropriate  $\boldsymbol{\alpha}$  values can be utilized to greatly reduce the computational burden of Eq. (17) and make the proposed modeling method B-ZTSK-FS become scalable for large-scale training datasets.

Therefore, the objective function  $J(\mathbf{X}, \mathbf{Z}, \mathbf{U}, \mathbf{Y}, \mathbf{p}_0)$  of the proposed likelihood model for zero-order TSK fuzzy system modeling indeed unifies clustering method for antecedent parameters of fuzzy rules and LS method for consequent parameters of fuzzy rules. That is to say, the parameter learning in  $J(\mathbf{X}, \mathbf{Z}, \mathbf{U}, \mathbf{Y}, \mathbf{p}_0)$  unifies

**Table 2**  
Sampling characteristic induced by Dirichlet distribution with different  $\alpha$  values.

| $\mathbf{u}_j$    | $\alpha = 0.001 \times \mathbf{1}_K$ |   |   |   |   | $\alpha = 0.01 \times \mathbf{1}_K$ |        |        |        |   | $\alpha = 0.1 \times \mathbf{1}_K$ |        |        |        |        |
|-------------------|--------------------------------------|---|---|---|---|-------------------------------------|--------|--------|--------|---|------------------------------------|--------|--------|--------|--------|
| $\mathbf{u}_1$    | 0                                    | 0 | 0 | 1 | 0 | 0                                   | 0      | 0      | 1      | 0 | 0.0177                             | 0.3105 | 0      | 0.6718 | 0      |
| $\mathbf{u}_2$    | 0                                    | 0 | 1 | 0 | 0 | 0                                   | 0      | 1      | 0      | 0 | 0                                  | 0      | 1      | 0      | 0      |
| $\mathbf{u}_3$    | 0                                    | 1 | 0 | 0 | 0 | 0                                   | 0      | 1      | 0      | 0 | 0.0002                             | 0.4306 | 0      | 0.5691 | 0.0002 |
| $\mathbf{u}_4$    | 0                                    | 0 | 0 | 0 | 1 | 0                                   | 1      | 0      | 0      | 0 | 0.3513                             | 0.0014 | 0.0001 | 0.0097 | 0.6375 |
| $\mathbf{u}_5$    | 1                                    | 0 | 0 | 0 | 0 | 0.0002                              | 0      | 0.9998 | 0      | 0 | 0.5376                             | 0      | 0.4288 | 0      | 0.0336 |
| $\mathbf{u}_6$    | 0                                    | 1 | 0 | 0 | 0 | 0                                   | 0      | 1      | 0      | 0 | 0.0264                             | 0.8803 | 0      | 0.0097 | 0.0836 |
| $\mathbf{u}_7$    | 0                                    | 0 | 0 | 0 | 1 | 0                                   | 1      | 0      | 0      | 0 | 0.1614                             | 0      | 0.8298 | 0.0035 | 0.0054 |
| $\mathbf{u}_8$    | 0                                    | 0 | 0 | 0 | 1 | 0                                   | 0.9999 | 0      | 0.0001 | 0 | 0.2556                             | 0.3665 | 0      | 0.2280 | 0.1499 |
| $\mathbf{u}_9$    | 0                                    | 1 | 0 | 0 | 0 | 0                                   | 0.4272 | 0      | 0.5728 | 0 | 0.9851                             | 0      | 0.0149 | 0      | 0      |
| $\mathbf{u}_{10}$ | 0                                    | 0 | 0 | 1 | 0 | 0                                   | 1      | 0      | 0      | 0 | 0.9956                             | 0.0012 | 0      | 0.0027 | 0.0004 |

the antecedent parameters learning and the consequent parameters learning in zero-order TSK fuzzy system modeling, thus both antecedent parameters and consequent parameters can be learnt simultaneously.

**Remark 2.** Most of the existing TSK fuzzy system modeling methods, including L1-norm penalty-based  $\epsilon$ -insensitive training method (LSSLI) [45] and L2-norm penalty-based  $\epsilon$ -insensitive TSK fuzzy model (L2-TSK-FS) [40], require that antecedent parameters of fuzzy rules have to be determined in advance such that they become not tunable later on, thus only free consequent parameters in each fuzzy rule need to be tuned. In other words, these antecedent parameters are determined without considering the intrinsic link between the distributions of both input data and output data, so the interpretability obtained may not be good enough. Another shortcoming of this kind of methods is its non-scalability for a large-scale training dataset. However, the proposed joint likelihood model here combines antecedent parameter learning together with consequent parameter learning by unifying two isolated objective functions that only separately consider either consequent part or antecedent part. This is to indicate that both antecedent and consequent parameters can be optimized simultaneously when the joint likelihood trends toward the maximization such that the interpretability about antecedent and consequent parameters of fuzzy rules can be obtained. What is more, the introduction of Bayesian inference framework can help us achieve the construction of zero-order TSK fuzzy system from the probabilistic perspective, and the introduction of the Dirichlet prior distribution can help us realize zero-order TSK fuzzy system modeling through iterative sampling technique and even make the construction of zero-order TSK fuzzy system feasible for large-scale training datasets by appropriately setting the Dirichlet index.

Please note, theoretically, maximum likelihood estimation (MLE) can deal with parameter estimation for any likelihood function. However, due to too many parameters  $u_j, y_k, p_0^k$  in  $J(\mathbf{X}, \mathbf{Z}, \mathbf{U}, \mathbf{Y}, \mathbf{p}_0)$ , we can hardly estimate them simultaneously using MLE. To simplify the solution process, we only estimate one parameter vector at a time, and other parameters are fixed. Thus, for the fixed  $\mathbf{Y}, \mathbf{p}_0$ , the joint likelihood is formulated as the following posteriori probability model:

$$p(\mathbf{X}, \mathbf{Z}, \mathbf{U}|\mathbf{Y}, \mathbf{p}_0) = p(\mathbf{X}, \mathbf{Z}|\mathbf{U}, \mathbf{Y}, \mathbf{p}_0)p(\mathbf{U}|\mathbf{Y}, \mathbf{p}_0) \\ \propto \exp \left\{ -\frac{1}{2} \sum_{j=1}^N \sum_{k=1}^K u_{jk}^m \|(\mathbf{x}_j^T, z_j) - (\mathbf{y}_k^T, p_0^k)\|^2 \right\} \times \left( \prod_{j=1}^N \prod_{k=1}^K u_{jk}^{\alpha_k-1} \right). \quad (20)$$

Next, in order to estimate  $u_j$ , for the fixed other parameters, Eq. (20) is simplified to the following posteriori probability model:

$$p(\mathbf{x}_j, z_j, u_j|\mathbf{Y}, \mathbf{p}_0) = p(\mathbf{x}_j, z_j|u_j, \mathbf{Y}, \mathbf{p}_0)p(u_j|\mathbf{Y}, \mathbf{p}_0) \\ \propto \prod_{k=1}^K \exp \left\{ -\frac{1}{2} u_{jk}^m \|(\mathbf{x}_j^T, z_j) - (\mathbf{y}_k^T, p_0^k)\|^2 \right\} u_{jk}^{\alpha_k-1}. \quad (21)$$

Thus the MLE problem is formulated as the MAP problem for parameter estimation based on the known prior probability.

Similarly, in order to estimate  $y_k, p_0^k$ , for the fixed other parameters, a posteriori probability model can be formed from the joint likelihood model  $p(\mathbf{X}, \mathbf{Z}, \mathbf{U}, \mathbf{Y}, \mathbf{p}_0)$  through Bayesian inference.

$$p(\mathbf{X}, \mathbf{Z}, y_k, p_0^k|\mathbf{U}) = p(\mathbf{X}, \mathbf{Z}|\mathbf{U}, y_k, p_0^k) p(y_k, p_0^k) \\ \propto \exp \left\{ -\frac{1}{2} \sum_{j=1}^N u_{jk}^m \|(\mathbf{x}_j^T, z_j) - (\mathbf{y}_k^T, p_0^k)\|^2 \right\} \\ \times \exp \left\{ -\frac{1}{2} \sum_{j=1}^N \sum_{k=1}^K \left( \tilde{\mu}^k(\mathbf{x}_j) \cdot p_0^k - z_j \right)^2 \right\}. \quad (22)$$

### 3.2. Parameters learning

In this subsection, the primary task we are concerned with is finding the MAP values of the parameters in the posterior probability model given the data objects. This is equivalent to finding the globally optimal values of the parameters. To perform the MAP inference, we use a Metropolis-Hastings sampling technique [22] to leverage its optimality guarantees. The corresponding parameter learning of zero-order TSK fuzzy system by the proposed modeling method B-ZTSK-FS is illustrated in Fig. 2. In terms of Fig. 2, we summarize the proposed learning method B-ZTSK-FS as follows:

**Step 1:** Define and initialize the parameters of B-ZTSK-FS.

- 1) We hereby assume that the cluster centers are in line with the Gaussian distribution  $\mathcal{N}(\boldsymbol{\mu}_y, \sum \mathbf{y})$ , so the parameters in this distribution can be initially set to the mean of the data objects in the empirical Bayes way,

$$\boldsymbol{\mu}_y = \frac{1}{N} \sum_{j=1}^N (\mathbf{x}_j^T, z_j)^T, \quad (23)$$

and a wide covariance in the shape of the covariance of the data objects,

$$\sum \mathbf{y} = \frac{\gamma}{N} \sum_{j=1}^N \left( (\mathbf{x}_j^T, z_j)^T - \boldsymbol{\mu}_y \right) \left( (\mathbf{x}_j^T, z_j)^T - \boldsymbol{\mu}_y \right)^T, \quad (24)$$

where  $\gamma$  is a preset parameter affecting the strength of the prior. In terms of our extensive experiments, we appropriately set  $\gamma = 3$ .

- 2) Initialize the fuzzy memberships  $u_j$  for all  $j = 1, 2, \dots, N$ , using a Dirichlet prior distribution in Eq. (25). For the value of  $\alpha$ , we experimentally set  $\alpha$  to  $\mathbf{1}_K$  for a small dataset ( $N \leq 10^3$ ), and set  $\alpha$  to  $0.1 \times \mathbf{1}_K$  for a medium dataset ( $10^3 < N \leq 10^4$ ), and set  $\alpha$  to  $0.01 \times \mathbf{1}_K$  for a comparatively large dataset ( $10^4 < N \leq 10^5$ ), and set  $\alpha$  to  $0.001 \times \mathbf{1}_K$  for a large dataset ( $N > 10^5$ ). According to our

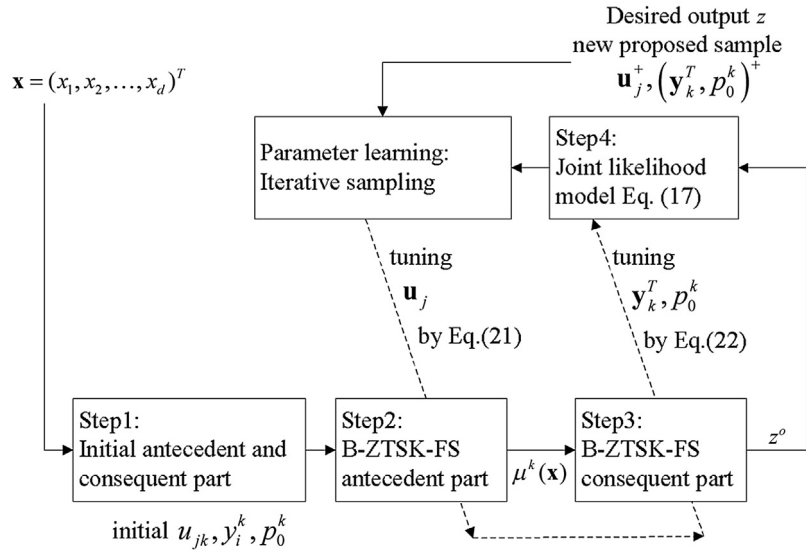


Fig. 2. Parameter learning in the proposed modeling method.

experiments, these four different  $\alpha$  values actually mean the corresponding sampling rates, i.e., about 100%, 70%, 10%, and 1% non-zero fuzzy membership values among all fuzzy membership values generated by the Dirichlet prior distribution with these four different  $\alpha$  values.

$$\mathbf{u}_j^+ \sim \text{Dirichlet}(\alpha) \quad (25)$$

- 3) Initialize the cluster centers  $(\mathbf{y}_k^T, p_0^k)$  using  $\mathcal{N}(\boldsymbol{\mu}_y, \sum \mathbf{y})$  for all  $k = 1, 2, \dots, K$ .
- 4) Set the MAP samples to current  $\mathbf{u}_j^*$  and  $(\mathbf{y}_k^T, p_0^k)^*$  using  $\mathbf{u}_j$  and  $(\mathbf{y}_k^T, p_0^k)$  respectively, for all  $j = 1, 2, \dots, N, k = 1, 2, \dots, K$ .

**Step 2:** Sample the fuzzy membership vectors using a Metropolis-Hastings sampling step given the input-output data  $\mathbf{X}, \mathbf{Z}$ , and cluster centers matrix  $(\mathbf{Y}, \mathbf{p}_0)$ .

- 1) Sample the proposed new membership vector  $\mathbf{u}_j^+$  using Eq. (25).
- 2) If  $\mathbf{u}_j^+$  increases the value of Eq. (21), then replaces the previous  $\mathbf{u}_j^*$  with  $\mathbf{u}_j^+$ .
- 3) In addition, in order to speed up the search for a good answer, and there is leaving open the probability of finding a new full MAP sample as the sampler progresses. Each proposed new membership sample  $\mathbf{u}_j^+$  is accepted to replace  $\mathbf{u}_j$  according to the probability,

$$a_{\mathbf{u}} = \min \left\{ 1, \frac{p(\mathbf{x}_j, z_j, \mathbf{u}_j^+ | \mathbf{Y}, \mathbf{p}_0)}{p(\mathbf{x}_j, z_j, \mathbf{u}_j | \mathbf{Y}, \mathbf{p}_0)} \right\}. \quad (26)$$

**Step 3:** Sample the cluster centers using a Metropolis-Hastings sampling step given the input-output data  $\mathbf{X}, \mathbf{Z}$ , and the membership matrix  $\mathbf{U}$ .

- 1) Sample the proposed new cluster center vector  $(\mathbf{y}_k^T, p_0^k)^+$  using  $\mathcal{N}(\mathbf{y}_k, \sum \mathbf{y})/\delta$ , where  $\delta$  is a user-set parameters that controls the tightness of the proposal around the current state of Metropolis-Hastings and relates to the vector acceptance rate (i.e., the percentage of accepting the proposed new cluster center vec-

tor). In terms of our extensive experiments, we appropriately set  $\delta = 10$ .

- 2) If  $(\mathbf{y}_k^T, p_0^k)^+$  increases the value of Eq. (22), then replaces the previous  $(\mathbf{y}_k^T, p_0^k)^*$  with  $(\mathbf{y}_k^T, p_0^k)^+$ .
- 3) In addition, in order to speed up the search for a good answer, and there is also leaving open the probability of finding a new MAP sample as the sampler progresses. A proposed new cluster center sample  $(\mathbf{y}_k^T, p_0^k)^+$  is accepted to replace  $(\mathbf{y}_k^T, p_0^k)$  according to the probability,

$$a_{\mathbf{y}} = \min \left\{ 1, \frac{p(\mathbf{X}, \mathbf{Z}, (\mathbf{y}_k^T, p_0^k)^+ | \mathbf{U})}{p(\mathbf{X}, \mathbf{Z}, (\mathbf{y}_k^T, p_0^k) | \mathbf{U})} \right\}. \quad (27)$$

**Step 4:** Check the samples used by the maximum joint likelihood in Eq. (17). After each sample  $\{\mathbf{U}, (\mathbf{Y}, \mathbf{p}_0)\}$  is generated by Eqs. (26) and (27), its likelihood is compared against the current MAP sample  $\{\mathbf{U}^*, (\mathbf{Y}, \mathbf{p}_0)^*\}$  generated by Eq. (21) and Eq. (22) in terms of Eq. (17), and is retained as the new MAP sample if it is an improvement for the joint likelihood in Eq. (17).

**Step 5:** Repeat step 2, step 3 and step 4, until the preset number  $N_{iter}$  of iterations is achieved.

**Remark 3.** The above modeling method B-ZTSK-FS contains two different sampling procedures: 1) The Metropolis-Hastings sampling technique is used to sample both fuzzy cluster membership vectors and cluster centers; 2) For datasets with different sizes, different  $\alpha$  values may be taken to make the corresponding Dirichlet distribution sample/generate different numbers of non-zero fuzzy cluster membership values. In this way, the computational burden of Eqs. (21) and (22) can be greatly reduced, especially for large-scale datasets.

### 3.3. Prediction

Once the above parameter learning is completed, the proposed modeling method B-ZTSK-FS is ready for prediction to a new input data object  $\mathbf{x} \in \mathbb{R}^d$ . The prediction procedure can be summarized as follows:

**Step 1:** Calculate the membership functions in Eq. (2) for all attributes  $x_i$  of every fuzzy rule using the learnt parameters  $y_i^k$  and  $u_{jk}$ , in which  $\delta_i^k$  can be calculated using Eq. (7).

**Step 2:** Compute the firing strength of every fuzzy rule by using Eq. (3).

**Step 3:** Compute the output function of every fuzzy rule  $z^k(\mathbf{x})$  using Eq. (4).

**Step 4:** Compute the output for the new input data object  $\mathbf{x}$  using Eq. (5), thus the predicted output  $z^0$  of the new input data object  $\mathbf{x}$  have been obtained by this time.

### 3.4. On fuzzy index and Dirichlet index

The proposed modeling method B-ZTSK-FS has two parameters that must be set before running. They are the fuzzy index  $m$  and the Dirichlet index  $\alpha$ . Below we will discuss how to set them in our experiments.

- 1) Dirichlet index  $\alpha$ : The parameter  $\alpha$  controls the shape of the Dirichlet prior distribution on the fuzzy membership values, and should be greater than zero. When  $\alpha$ 's elements are less than one, fuzzy membership values induced by the corresponding Dirichlet prior distribution tend towards binary values (i.e., 0 or 1). When  $\alpha$ 's elements are greater than one, more non-zero fuzzy membership values will be yielded. In practice, in order to make the proposed method B-ZTSK-FS suitable for training datasets with different sizes, we should appropriately set  $\alpha$ . In terms of our extensive experiments, here we give a guideline about its values. We manually set  $\alpha$  to be the  $K$  dimensional unit vector  $\mathbf{1}_K$  for a small dataset ( $N \leq 10^3$ ), and set  $\alpha$  to  $0.1 \times \mathbf{1}_K$  for a medium dataset ( $10^3 < N \leq 10^4$ ), and set  $\alpha$  to  $0.01 \times \mathbf{1}_K$  for a comparatively large dataset ( $10^4 < N \leq 10^5$ ), and set  $\alpha$  to  $0.001 \times \mathbf{1}_K$  for a large dataset ( $N > 10^5$ ).
- 2) Fuzzy index  $m$ : This index indeed plays the same role as in the standard FCM algorithm [35] in which fuzzy cluster membership values become more crisp as  $m$  approaches 1. Although  $m$  is data dependent for the proposed modeling method B-ZTSK-FS, our comprehensive experiments indicate that  $m$  being 2 is an appropriate choice for the proposed modeling method B-ZTSK-FS.

### 3.5. On time complexity of B-ZTSK-FS

Here we mainly focus on the time complexity of the proposed zero-order TSK fuzzy modeling method B-ZTSK-FS in training. Obviously, it mainly consists of two parts: the time complexity of searching MAP sample  $\mathbf{U}^*$  and the time complexity of searching MAP samples  $(\mathbf{Y}, \mathbf{p}_0)^*$ . Hence the asymptotic computational complexity of B-ZTSK-FS for once outer iteration is  $O(NK(d+1) + NK^2d)$ , where  $N$  is the number of data objects,  $d$  is the dimensionality of data,  $K$  is the number of clusters (i.e., fuzzy rules).  $O(NK(d+1))$  comes from Eq. (21) for searching MAP sample  $\mathbf{U}^*$ ;  $O(NK^2d)$  comes from Eq. (22) for searching MAP sample  $(\mathbf{Y}, \mathbf{p}_0)^*$ .

In addition, we have experimentally found that for low dimensional (e.g.  $d \leq 6$ ) datasets with few clusters (i.e., the number of rules), a small number  $N_{iter}$  of iterations being 100 is generally sufficient, while for high dimensional (e.g.  $d > 6$ ) datasets with many clusters, a larger  $N_{iter}$  being 1000 is often needed.

## 4. Experimental results

In order to analyze the performance of the proposed modeling method B-ZTSK-FS, we have used 28 synthetic and real-world datasets with different numbers of attributes and instances from the KEEL project repository [46]. Table 3 shows the main characteristics of the datasets, with the number of input attributes ranging from 2 to 40, and the number of instances from 337 to 40,768. The most complex datasets are the ones in the last 7 rows (Table 3), due to both the large number of attributes and instances.

**Table 3**  
Datasets adopted for the experimental study.

| Datasets                      | Abbreviation | Attributes | Instances |
|-------------------------------|--------------|------------|-----------|
| Electrical Length             | ELE1         | 2          | 495       |
| Plastic Strength              | PLA          | 2          | 1650      |
| Quake                         | QUA          | 3          | 2178      |
| Electrical Maintenance        | ELE2         | 4          | 1056      |
| Friedman                      | FRIE         | 5          | 1200      |
| Auto MPG6                     | MPG6         | 5          | 392       |
| Delta Ailerons                | DELA1L       | 5          | 7129      |
| Daily Electricity Energy      | DEE          | 6          | 365       |
| Delta Elevators               | DELELV       | 6          | 9517      |
| Analcatt                      | ANA          | 7          | 4052      |
| Auto MPG8                     | MPG8         | 7          | 398       |
| Abalone                       | ABA          | 8          | 4177      |
| California Housing            | CAL          | 8          | 20640     |
| Concrete Compressive Strength | CON          | 8          | 1030      |
| Stock prices                  | STP          | 9          | 950       |
| Weather Ankara                | WAN          | 9          | 1609      |
| Weather Izmir                 | WIZ          | 9          | 1461      |
| Forest Fires                  | FOR          | 12         | 517       |
| Mortgage                      | MOR          | 15         | 1049      |
| Treasury                      | TRE          | 15         | 1049      |
| Baseball                      | BAS          | 16         | 337       |
| MV Artificial Domain          | MV           | 10         | 40768     |
| House-16H                     | HOU          | 16         | 22784     |
| Elevators                     | ELV          | 18         | 16559     |
| Computer Activity             | CA           | 21         | 8192      |
| Pole Telecommunications       | POLE         | 26         | 14998     |
| Pumadyn                       | PUM          | 32         | 8192      |
| Ailerons                      | AIL          | 40         | 13750     |

This section is organized as follows. In Section 4.1, the experimental setup is described. The experimental results of the proposed modeling method and three well-developed fuzzy system modeling methods are reported in Section 4.2. Finally, we show the running time of the comparison methods on 28 datasets and discuss the scalability of the proposed method in Section 4.3.

### 4.1. Experimental setup

- 1) *Methods for Comparison*: Three comparison methods are the recently-developed L2-norm penalty-based e-insensitive TSK fuzzy modeling method (L2-TSK-FS) [40] and two evolutionary fuzzy modeling methods (LEL-TSK [47] and METSK-HD<sup>e</sup> [48]).

L2-TSK-FS [40] is a L2-norm penalty-based e-insensitive TSK fuzzy modeling method, which learns antecedent parameters and consequent parameters of fuzzy rules, separately. Antecedent parameters are learnt by the FCM algorithm and consequent parameters are learnt using L2-SVR that is solved by the quadratic programming.

LEL-TSK [47] is a two-stage local evolutionary learning method of TSK fuzzy rules based on MOGUL (a methodology to obtain genetic fuzzy rule-based systems under the iterative rule learning algorithm), which can obtain accurate local semantics-based TSK fuzzy rules by using the local fuzzy prototypes.

METSK-HD<sup>e</sup> [48] is a multi-objective evolutionary learning method of TSK fuzzy rules, which can learn an accurate TSK-fuzzy systems for the high-dimensional and large-scale regression problem.

- 2) *Parameter Settings*: The default values of all parameters used in L2-TSK-FS, LEL-TSK, and METSK-HD<sup>e</sup> are taken [40,48,49], as listed in Table 4. In addition, for other non-default parameters, unless specified, the five-fold cross validation strategy is adopted to determine the optimal setting within the given grids. The search grids for cross validation are also listed in Table 4.



**Table 4**  
Parameters in the four methods.

| Methods                    | Parameters                                                  | values                                              |
|----------------------------|-------------------------------------------------------------|-----------------------------------------------------|
| L2-TSK-FS [40]             | The number $K$ of fuzzy rules                               | $K \in \{4, 9, 16, 25, 36, 49, 64, 81, 100, 121\}$  |
|                            | Regularization parameter                                    | $t \in \{10^{-12}, \dots, 10^0, \dots, 10^{12}\}$ . |
| LEL-TSK [47]               | Population size                                             | 61                                                  |
|                            | Crossover probability                                       | 0.6                                                 |
|                            | Mutation probability                                        | 0.1                                                 |
| METSK-HD <sup>c</sup> [48] | Population size                                             | 200                                                 |
|                            | External population size                                    | 61                                                  |
|                            | Maximum number of evaluations                               | 100,000                                             |
|                            | Mutation probability                                        | 0.2                                                 |
|                            | Number of labels                                            | $\{2, \dots, 7\}$                                   |
|                            | Bits per gene for the gray codification                     | 30                                                  |
|                            | Rate of examples used to estimate the error                 | 0.2                                                 |
|                            | Maximum of examples used to estimate the error              | 1000                                                |
|                            | Maximum number of rules for the rule cropping (first stage) | 100                                                 |
| B-ZTSK-FS                  | The number $K$ of fuzzy rules                               | $K \in \{4, 9, 16, 25, 36, 49, 64, 81, 100, 121\}$  |
|                            | The fuzzy index $m$                                         | 2                                                   |
|                            | Dirichlet index $\alpha$ is fixed according Section 3.4     | See Section 3.4                                     |

**Table 5**  
Average results of the four methods.

| Datasets (Attributes/Instances) | L2-TSK-FS |              |              | LEL-TSK |              |              | METSK-HD <sup>c</sup> |              |              | B-ZTSK-FS |              |               |
|---------------------------------|-----------|--------------|--------------|---------|--------------|--------------|-----------------------|--------------|--------------|-----------|--------------|---------------|
|                                 | $R$       | $MSE_{tra}$  | $MSE_{tst}$  | $R$     | $MSE_{tra}$  | $MSE_{tst}$  | $R$                   | $MSE_{tra}$  | $MSE_{tst}$  | $R$       | $MSE_{tra}$  | $MSE_{tst}$   |
| ELE1(2/495)                     | 25        | 1.416        | 2.073        | 27      | <b>1.190</b> | 2.402        | <b>11.4</b>           | 1.350        | <b>2.022</b> | 16        | 1.684        | 2.137         |
| PLA(2/1650)                     | 49        | 1.046        | 1.208        | 66      | 1.032        | 1.188        | 19.2                  | 1.057        | 1.136        | <b>16</b> | <b>1.013</b> | <b>1.065</b>  |
| QUA(3/2178)                     | 36        | 0.014        | 0.025        | 127     | 0.015        | 0.031        | 18.3                  | 0.017        | 0.018        | <b>16</b> | <b>0.011</b> | <b>0.013</b>  |
| ELE2(4/1056)                    | <b>9</b>  | 7682         | 7979         | 44.8    | 2928         | 3752         | 36.9                  | 2270         | 3192         | 9         | <b>2084</b>  | <b>2909</b>   |
| FRIE(5/1200)                    | 49        | 0.796        | 2.956        | 435     | <b>0.322</b> | <b>1.070</b> | 66                    | 1.075        | 1.888        | <b>25</b> | 0.967        | 1.717         |
| MPG6(5/392)                     | 49        | 1.202        | 5.562        | 79      | 1.473        | 6.357        | 53.6                  | 1.082        | 4.478        | <b>36</b> | <b>1.016</b> | <b>3.201</b>  |
| DELAIR(5/7129)                  | –         | –            | –            | 105.2   | 1.193        | 1.760        | 36.8                  | <b>1.190</b> | <b>1.402</b> | <b>25</b> | 1.212        | 1.533         |
| DEE(6/365)                      | 36        | 0.195        | 0.289        | 57.8    | 0.662        | 0.682        | 50.6                  | 0.030        | 0.103        | <b>36</b> | <b>0.028</b> | <b>0.087</b>  |
| DELELV(6/9517)                  | –         | –            | –            | 219.9   | <b>0.964</b> | 2.788        | 39.1                  | 0.972        | <b>1.031</b> | <b>36</b> | 1.009        | 1.212         |
| ANA(7/4052)                     | <b>25</b> | 0.025        | 0.027        | 301     | 0.009        | 0.014        | 33.3                  | 0.002        | 0.004        | 36        | <b>0.002</b> | <b>0.003</b>  |
| MPG8(7/398)                     | 64        | <b>1.135</b> | <b>4.763</b> | 129     | 1.574        | 7.111        | 64.2                  | 1.154        | 5.391        | <b>25</b> | 1.231        | 8.744         |
| ABA(8/4177)                     | 25        | <b>1.879</b> | 2.558        | 107     | 2.040        | 2.412        | 23.1                  | 2.205        | 2.392        | <b>9</b>  | 2.063        | <b>2.180</b>  |
| CAL(8/20640)                    | –         | –            | –            | 542     | 2.370        | 2.540        | 55.8                  | <b>1.640</b> | <b>1.710</b> | <b>9</b>  | 1.766        | 1.810         |
| CON(8/1030)                     | 49        | 14.50        | 26.24        | 325     | <b>10.69</b> | 31.43        | 53.7                  | 15.05        | 23.89        | <b>36</b> | 11.72        | <b>18.59</b>  |
| STP(9/950)                      | 36        | <b>0.125</b> | <b>0.287</b> | 78.9    | 0.606        | 0.849        | 66.4                  | 0.167        | 0.387        | <b>36</b> | 0.177        | 0.310         |
| WAN(9/1609)                     | 25        | 0.698        | 0.891        | 123     | 0.709        | 1.632        | 48                    | 0.701        | 1.189        | <b>16</b> | <b>0.683</b> | <b>0.797</b>  |
| WIZ(9/1461)                     | <b>16</b> | <b>0.509</b> | <b>0.656</b> | 116     | 0.699        | 2.227        | 29.1                  | 0.729        | 0.944        | 25        | 0.533        | 0.691         |
| FOR(12/517)                     | 49        | 165.6        | 5651         | 418     | <b>160.4</b> | 14074        | 40.6                  | 551.4        | 5587         | <b>36</b> | 417.1        | <b>4667</b>   |
| MOR(15/1049)                    | 25        | <b>0.001</b> | <b>0.003</b> | 64.3    | 0.259        | 0.472        | 27.2                  | 0.005        | 0.013        | <b>16</b> | 0.006        | 0.014         |
| TRE(15/1049)                    | 16        | <b>0.008</b> | <b>0.018</b> | 63.7    | 0.267        | 0.504        | 28.1                  | 0.017        | 0.038        | <b>9</b>  | 0.013        | 0.025         |
| BAS(16/337)                     | 36        | 111909       | 1020075      | 374     | <b>9607</b>  | 461402       | 59.8                  | 47900        | 368820       | <b>25</b> | 45712        | <b>324107</b> |
| MV(10/40768)                    | –         | –            | –            | –       | –            | –            | 56.5                  | 0.060        | 0.061        | <b>36</b> | <b>0.052</b> | <b>0.055</b>  |
| HOU(16/22784)                   | –         | –            | –            | –       | –            | –            | 30.5                  | 8.290        | 8.640        | <b>25</b> | <b>6.542</b> | <b>7.109</b>  |
| ELV(18/16559)                   | –         | –            | –            | –       | –            | –            | 34.9                  | 6.750        | 7.021        | <b>25</b> | <b>5.905</b> | <b>6.372</b>  |
| CA(21/8192)                     | –         | –            | –            | –       | –            | –            | 32.91                 | 4.376        | 4.949        | <b>25</b> | <b>3.839</b> | <b>4.263</b>  |
| POLE(26/14998)                  | –         | –            | –            | –       | –            | –            | 46.3                  | <b>57.96</b> | <b>61.02</b> | <b>36</b> | 62.56        | 66.85         |
| PUM(32/8192)                    | –         | –            | –            | –       | –            | –            | 63.3                  | 0.267        | <b>0.287</b> | <b>36</b> | <b>0.253</b> | 0.324         |
| AIL(40/13750)                   | –         | –            | –            | –       | –            | –            | 48.4                  | 1.390        | 1.510        | <b>36</b> | <b>1.279</b> | <b>1.333</b>  |

<sup>c</sup>The results ( $MSE_{tra}$  and  $MSE_{tst}$ ) on the ELE1, DELAIL, DELELV, CAL, HOU, ELV, PUM and AIL in this table should be multiplied by  $10^5, 10^{-8}, 10^{-6}, 10^9, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-8}$ , respectively.

3) *Evaluation index*: In order to evaluate the approximation performance of the modeling methods, the following performance index, mean squared error ( $MSE$ ) is used in the experiments [49],

$$MSE = \frac{1}{2 \cdot N} \sum_{j=1}^N (z_j^o - z_j)^2 \quad (28)$$

where  $N$  is the number of data objects,  $z_j$  is the desired output for the  $j$ th input data object,  $z_j^o$  is the predicted output of one fuzzy modeling method for the  $j$ th data object. The smaller the value of  $MSE$ , the better the approximation performance.

4) *Experimental Environment*: All the methods adopted in this paper were implemented with the MATLAB codes on a computer with an Intel Processor i3 M 370 2.4 GHz, with 6 GB memory.

In the following sections, we will report the average experimental results of all the comparison methods obtained by five-fold cross validation strategy. For every run, each adopted dataset is randomly split into 5 equal parts, each containing its 20% instances, 4 parts being for training and the remaining one being for testing. For each of the five partitions, six trials (6 different seeds) of the methods are executed. So, we obtain the average results for 30 runs for each adopted dataset.

#### 4.2. On accuracy and interpretability

Table 5 shows the average results of B-ZTSK-FS and other three methods for comparison, in which three indices, i.e., the average number of rules ( $R$ ), the average  $MSE$ s respectively in training and test data (i.e.,  $MSE_{tra}$ ,  $MSE_{tst}$ ), are considered. The best values of

**Table 6**

Five fuzzy rules obtained by B-ZTSK-FS on the Friedman dataset.

| TSK Fuzzy Rule $R^k$ :                                                                                                                                                                                                               |                                                                                                                        |                       |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------|
| IF $x_1$ is $A_1^k(y_1^k, \delta_1^k) \wedge x_2$ is $A_2^k(y_2^k, \delta_2^k) \wedge x_3$ is $A_3^k(y_3^k, \delta_3^k) \wedge x_4$ is $A_4^k(y_4^k, \delta_4^k) \wedge x_5$ is $A_5^k(y_5^k, \delta_5^k)$ , THEN $f^k(x) = p_0^k$ . |                                                                                                                        |                       |                 |
| No. of rules                                                                                                                                                                                                                         | Antecedent parameters                                                                                                  | Consequent parameters | Desired outputs |
| $k$                                                                                                                                                                                                                                  | $\mathbf{y}^k = (y_1^k, y_2^k, \dots, y_d^k)^T, \delta^k = (\delta_1^k, \delta_2^k, \dots, \delta_d^k)^T$              | $p_0^k$               | $z$             |
| 1                                                                                                                                                                                                                                    | $\mathbf{y}^1 = [0.6673, 0.6843, 0.5147, 0.8965, 0.6980]^T$<br>$\delta^1 = [0.0027, 0.0038, 0.0131, 0.0025, 0.0053]^T$ | $p_0^1 = 22.3667$     | 22.3667         |
| 2                                                                                                                                                                                                                                    | $\mathbf{y}^2 = [0.3366, 0.4069, 0.4520, 0.2794, 0.4436]^T$<br>$\delta^2 = [0.0052, 0.0066, 0.0052, 0.0039, 0.0052]^T$ | $p_0^2 = 9.2297$      | 9.2293          |
| 3                                                                                                                                                                                                                                    | $\mathbf{y}^3 = [0.4814, 0.4866, 0.5141, 0.4449, 0.4814]^T$<br>$\delta^3 = [0.0070, 0.0069, 0.0058, 0.0053, 0.0053]^T$ | $p_0^3 = 13.5728$     | 13.5726         |
| 4                                                                                                                                                                                                                                    | $\mathbf{y}^4 = [0.6273, 0.4741, 0.5629, 0.5941, 0.5274]^T$<br>$\delta^4 = [0.0056, 0.0058, 0.0066, 0.0046, 0.0059]^T$ | $p_0^4 = 16.6991$     | 16.6991         |
| 5                                                                                                                                                                                                                                    | $\mathbf{y}^5 = [0.3020, 0.2733, 0.4768, 0.1833, 0.3436]^T$<br>$\delta^5 = [0.0054, 0.0060, 0.0039, 0.0025, 0.0041]^T$ | $p_0^5 = 6.1254$      | 6.1258          |

three indices are marked in bold in Table 5. Besides, since L2-TSK-FS and LEL-TSK can not work under our experimental environment for a large number of attributes and/or instances, we do not report the corresponding results (denoted by “–”) for several datasets in Table 5.

It can be readily seen from this table that among 28 datasets, B-ZTSK-FS wins 16, METSK-HD<sup>e</sup> wins 6, L2-TSK-FS wins 5 and LEL-TSK wins only once, in the sense of approximation accuracy. In the sense of the complexity of the model, the proposed method B-ZTSK-FS obtains the smallest number of rules on 24 datasets, L2-TSK-FS smallest on 3 datasets, METSK-HD<sup>e</sup> smallest on one datasets.

Now, let us observe the interpretability of the obtained fuzzy rules. Due to the limitation of the paper's space, only five fuzzy rules obtained by the proposed method for the Friedman dataset are illustrated in Table 6. Given an input, we can compute the desired output of the Friedman function. Therefore, we can readily verify the relationship between antecedent/consequent parameters of fuzzy rules and desired output. That is to say, with the antecedent parameters of a fuzzy rule, i.e.,  $\mathbf{y}^k$ , as the input, the values of the consequent parameter is close to the corresponding desired output, which is in line with the claim about the interpretability in subsection 3.1.

In order to analyze the statistical significance of these results obtained by the four comparison methods, here we report the experimental results about two non-parametric statistical tests [50,51], i.e., Friedman's test [52] for multiple comparison and Wilcoxon's test [53,54] for pair-wise comparison, such that we can rank these methods and check whether the difference between B-ZTSK-FS and each of the other comparison methods is statistically significant in terms of the test error  $MSE_{\text{test}}$ . In these two statistical tests, we use a level of confidence  $\alpha = 0.05$ . Because L2-TSK-FS and LEL-TSK do not work for several datasets, we observe the statistical significance of the experimental results from three cases.

**Case 1.** The four methods on the first 18 datasets.

Table 7 shows the rankings and p-values of these four methods by using Friedman's test, which actually indicates that the proposed method has the best ranking. Table 8 shows the results by using Wilcoxon's test, i.e., the test rejects the null hypothesis at a level of confidence  $\alpha = 0.05$ , and the p-values indicate that the difference is statistically significant. The above two results demonstrate that B-ZTSK-FS indeed outperforms the other methods in the sense of  $MSE_{\text{test}}$ .

**Case 2.** The three methods on the first 21 datasets.

Both Table 9 about Friedman's test and Table 10 about Wilcoxon's test clearly indicate the same claim as in the above for three methods LEL-TSK, METSK-HD<sup>e</sup> and B-ZTSK-FS.

**Case 3.** The two methods on all the datasets.

**Table 7**

Rankings and p-value obtained through Friedman's test (the first 18 datasets).

| Methods               | Ranking | p-value |
|-----------------------|---------|---------|
| B-ZTSK-FS             | 1.611   | <1E–4   |
| METSK-HD <sup>e</sup> | 2.222   |         |
| L2-TSK-FS             | 2.667   |         |
| LEL-TSK               | 3.500   |         |

**Table 8**

Results on Wilcoxon's test (the first 18 datasets).

| Comparison                          | Hypothesis | p-value |
|-------------------------------------|------------|---------|
| B-ZTSK-FS vs. METSK-HD <sup>e</sup> | Rejected   | 0.007   |
| B-ZTSK-FS vs. L2-TSK-FS             | Rejected   | 0.025   |
| B-ZTSK-FS vs. LEL-TSK               | Rejected   | 0.006   |

**Table 9**

Rankings and p-value obtained through Friedman's test (the first 21 datasets).

| Methods               | Ranking | p-value |
|-----------------------|---------|---------|
| B-ZTSK-FS             | 1.381   | <1E–5   |
| METSK-HD <sup>e</sup> | 1.762   |         |
| LEL-TSK               | 2.857   |         |

**Table 10**

Results on Wilcoxon's test (the first 21 datasets).

| Comparison                          | Hypothesis | p-value |
|-------------------------------------|------------|---------|
| B-ZTSK-FS vs. METSK-HD <sup>e</sup> | Rejected   | 0.046   |
| B-ZTSK-FS vs. LEL-TSK               | Rejected   | 0.002   |

**Table 11**

Rankings and p-value obtained through Friedman's test (all 28 datasets).

| Methods               | Ranking | p-value |
|-----------------------|---------|---------|
| B-ZTSK-FS             | 1.286   | 0.020   |
| METSK-HD <sup>e</sup> | 1.714   |         |

Both Table 11 about Friedman's test and Table 12 about Wilcoxon's test clearly indicate the same claim as in the above for two methods METSK-HD<sup>e</sup> and B-ZTSK-FS.

In summary, by analyzing Tables 5–12, we can draw the following conclusions:

**Table 12**

Results on Wilcoxon's test (all 28 datasets).

| Comparison                          | Hypothesis | p-value |
|-------------------------------------|------------|---------|
| B-ZTSK-FS vs. METSK-HD <sup>e</sup> | Rejected   | 0.026   |

**Table 13**

Average running time of the different methods – hours, minutes and seconds (h:m:s).

| Datasets<br>(Attributes/Instances) | L2-TSK-FS      | LEL-TSK        | METSK-HD <sup>e</sup> | B-ZTSK-FS      |
|------------------------------------|----------------|----------------|-----------------------|----------------|
| ELE1(2/495)                        | 0:00:58        | 0:01:29        | 0:00:42               | <b>0:00:27</b> |
| PLA(2/1650)                        | 0:35:48        | 0:13:36        | 0:01:59               | <b>0:01:51</b> |
| QUA(3/2178)                        | 1:22:21        | 0:36:57        | 0:05:22               | <b>0:03:09</b> |
| ELE2(4/1056)                       | 0:09:23        | 0:06:44        | 0:09:42               | <b>0:03:15</b> |
| FRIE(5/1200)                       | 0:13:46        | 1:08:05        | 0:28:54               | <b>0:08:57</b> |
| MPG6(5/392)                        | <b>0:00:30</b> | 0:09:38        | 0:20:40               | 0:04:15        |
| DELA1L(5/7129)                     | –              | 2:05:38        | 1:48:01               | <b>0:52:30</b> |
| DEE(6/365)                         | <b>0:00:23</b> | 0:05:03        | 0:15:38               | 0:03:48        |
| DELELV(6/9517)                     | –              | 8:06:59        | 1:14:54               | <b>0:49:15</b> |
| ANA(7/4052)                        | 8:50:14        | 4:18:50        | 0:29:12               | <b>0:18:21</b> |
| MPG8(7/398)                        | <b>0:00:57</b> | 0:19:42        | 0:20:19               | 0:04:12        |
| ABA(8/4177)                        | 9:40:50        | 2:05:59        | 0:24:10               | <b>0:17:54</b> |
| CAL(8/20,640)                      | –              | <b>0:14:40</b> | 3:12:58               | 1:13:21        |
| CON(8/1030)                        | 0:08:43        | 1:22:57        | 0:25:41               | <b>0:08:33</b> |
| STP(9/950)                         | <b>0:06:50</b> | 0:27:21        | 0:30:39               | 0:09:30        |
| WAN(9/1609)                        | 0:33:12        | 1:13:09        | 0:31:20               | <b>0:10:24</b> |
| WIZ(9/1461)                        | 0:24:51        | 1:02:48        | 0:16:03               | <b>0:12:36</b> |
| FOR(12/517)                        | <b>0:01:06</b> | 4:54:52        | 0:19:33               | 0:08:51        |
| MOR(15/1049)                       | 0:09:12        | 0:22:11        | <b>0:05:31</b>        | 0:11:00        |
| TRE(15/1049)                       | 0:11:52        | 0:35:13        | <b>0:07:52</b>        | 0:10:18        |
| BAS(16/337)                        | <b>0:00:18</b> | 1:31:06        | 0:37:03               | 0:13:15        |
| MV(10/40,768)                      | –              | –              | 2:28:20               | <b>0:55:09</b> |
| HOU(16/22,784)                     | –              | –              | 3:31:02               | <b>0:59:52</b> |
| ELV(18/16,559)                     | –              | –              | 2:23:36               | <b>0:34:28</b> |
| CA(21/8192)                        | –              | –              | 2:46:36               | <b>0:21:58</b> |
| POLE(26/14,998)                    | –              | –              | 3:12:06               | <b>0:43:15</b> |
| PUM(32/8192)                       | –              | –              | 1:31:33               | <b>0:48:22</b> |
| AIL(40/13,750)                     | –              | –              | 4:02:09               | <b>1:13:48</b> |

- 1) LEL-TSK obtains very accurate results on the training sets, but it usually has bad test errors, which shows the method is easy to trap in overfitting and results in poor generalization performance.
- 2) Both L2-TSK-FS and METSK-HD<sup>e</sup> are very competitive in the results on both training and testing sets. However, L2-TSK-FS is not suitable for large-scale datasets due to the use of the quadratic programming for learning consequent parameters of fuzzy rules. Although METSK-HD<sup>e</sup> is applicable for relatively large-scale datasets with relatively high dimensions (see the last 7 datasets in Table 5), it generates a relatively complex model (a relatively large number of rules) and hence degrades the interpretability for fuzzy rules.
- 3) Among the four methods, the proposed method B-ZTSK-FS provides relatively less number of interpretable fuzzy rules such that B-ZTSK-FS guarantees strong interpretability. What is more, this proposed method obtains the best accuracy for both small or medium-sized datasets and large-scale or high dimensional datasets in general, which is shown in not only Table 5, but also statistical tests.

#### 4.3. Computational time and scalability

In order to observe the scalability of the four methods, we report the average running time of the four methods in Table 13. Compared with L2-TSK-FS, LEL-TSK and METSK-HD<sup>e</sup>, B-ZTSK-FS has the least running time among the majority of datasets. Since L2-TSK-FS uses the quadratic programming to solve L2-SVR for learning the consequent parameters of fuzzy rules, its running time grows exponentially with the increase of the number of instances, and thus it isn't suitable for large-scale datasets. In view of LEL-TSK and METSK-HD<sup>e</sup>, since their multi-stage evolutionary learning strategy aggravates the computational burden, LEL-TSK can not work for large-scale or high dimensional datasets, and the running time of METSK-HD<sup>e</sup> also increases rapidly with the increase of the number of attributes and instances. However, for most datasets, and even

the most complex datasets, B-ZTSK-FS still works well. Therefore, B-ZTSK-FS have better scalability than the other three comparison methods.

## 5. Conclusion

The outstanding properties of both fuzzy and probabilistic thinking have made fuzzy-systems and Bayesian-theory based modeling methods extensively studied. However, up to date, in contrast to their respective studies, little attention has been paid on the methods of making effective collaboration with them to obtain better performance. As an attempt in this aspect, a novel zero-order TSK fuzzy modeling method called B-ZTSK-FS is proposed to demonstrate that fuzziness and probability can co-jointly work very well in this study. With the help of Bayesian inference framework together with Dirichlet prior distribution assumption for fuzzy cluster memberships, a joint likelihood about zero-order TSK fuzzy system modeling is built to make the construction of TSK fuzzy system become scalable for large-scale training datasets in a way of simultaneous learning for interpretable antecedent and consequent parameters through the proposed iterative sampling algorithm. The extensive experimental results on 28 synthetic and real-world datasets indicate that the proposed modeling method B-ZTSK-FS is superior to the other comparison methods in the sense of approximation accuracy, interpretability and scalability.

Since it is the first attempt to construct the fuzzy system model by collaborating TSK fuzzy system modeling with Bayesian inference, our preliminary work here should be extended to first order TSK fuzzy system modeling in near future. Moreover, there are many other issues worthy to be addressed in future. For example, in this study, the number of fuzzy rules in B-ZTSK-FS is fixed empirically by using the cross-validation strategy. A better way to determine an appropriate number of fuzzy rules seems to exploit the benefit of Bayesian inference framework in estimating the number of clusters as done in [30]. Another issue is how to develop an enhanced version of the proposed modeling method B-ZTSK-FS here such that it is effective for noisy datasets. These will be the focus of our future work.

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